

**MODERN ALGEBRA:
INTRODUCTION TO REPRESENTATION THEORY
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1. LECTURE 1: REPRESENTATION OF A GROUP

The reference for the first part of this course (representation theory of finite groups) is [\[Ser77\]](#).

1.1. What is a representation of a group? Needless to say, the notion of a *group* is one of the most fundamental and important concepts in mathematics. Therefore, it is a basic motivation to “understand” a given group (say, G) in some sense. However, when can we really say “OK, now we have understood this group G ”?

Because this is not a rigorous (mathematically formulated) problem, there is also no rigorous answer; rather, there should be a lot of answers.

For example, one can claim that it is enough to describe the multiplication table for all elements of a given group G . The multiplication law of a group G is nothing but the whole group structure of G , so it indeed characterizes the group G . In this sense, it should be a complete answer to the above problem. However, as you can imagine easily, describing the multiplication table is not always so practical to express the nature of the given group G .

Example 1.1. Let G be the group whose elements consist of a, b, c, d satisfying the following multiplication table:

$\cdot$	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

In fact, this group is just $\mathbb{Z}/4\mathbb{Z}$, where $0, 1, 2, 3 \in \mathbb{Z}/4\mathbb{Z}$ are denoted by a, b, c, d , respectively. It is obvious that this way of expressing $\mathbb{Z}/4\mathbb{Z}$ is just making the situation complicated!

Another possible answer should be to describe a set of generators and relations of a given group G . Indeed, this can be a complete “answer”, in the sense that a set of generators and relations completely characterizes the group G . However, for any given group G , a set of generators and relations may not be unique; sometimes, a bad choice of generators and relations does not make the structure of the given group G transparent. To see this, let us look at the following bad example:

Example 1.2. Let $G := \mathbb{Z}/4\mathbb{Z}$. This group can be written as

$$\langle x, y \mid x^2 = y, y^2 = 1, xy = yx \rangle.$$

Of course, the relation $x^2 = y$ reduces this to

$$\langle x \mid x^4 = 1 \rangle,$$

which should be the most standard presentation of $\mathbb{Z}/4\mathbb{Z}$.

These observations suggest that there can be many different ways to formulate what it means to “understand a group”. *Representation theory*, which we will study in this course, is one way to formulate the meaning of “understand a group”.

Here, we caution that although we understand everything about representations of a given group G , it does not mean that we understand everything about G . For example, knowing all about representation theory of G does not necessarily enable us to describe the multiplication table of nor a set of generators and relations of G . What is important is that the notion of representation appears in various areas of mathematics (or, even mathematical physics). Of course, representations of groups are interesting objects of study in their own right, but much of their value also comes from their usefulness and applications in many different areas of mathematics.

Now let us define the notion of a representation.

Definition 1.3. Let G be a group. A *representation* of G is a pair (π, V) of a $\mathbb{C}$ -vector space V and an action of G on V , i.e., a map

$$G \times V \rightarrow V: (g, v) \mapsto g \cdot v$$

such that the induced map

$$\pi: G \rightarrow \text{Aut}_{\mathbb{C}}(V): g \mapsto [\pi(g): v \mapsto g \cdot v]$$

is a group homomorphism.

It is often said that “a group is a set of symmetries”. If we follow this slogan, we can say that “a representation of a group G is a realization of the symmetries contained in G on a vector space”. To put it somewhat poetically, a representation of a group is a “shadow” it casts on a vector space.

- Remark 1.4.**
- (1) By convention (which is normal in representation theory), we sometimes write only one of π and V to mean a representation (π, V) of a group G . For example, we often say “let π be a representation of G ” or “let V be a representation of G ”.
 - (2) Somewhat confusingly, we may also use π to refer to both the action and the underlying vector space. For example, we often say “let π be a representation of G and we take an element $v \in \pi$ ”.
 - (3) When we only write V for a representation of G , we often use the symbol “.” for denoting the action of G on V . For example, we write “let $g \in G$, $v \in V$ and we consider $g \cdot v$ ”.

For a given group G , there are various possible representations of G . For example, we always have the *trivial representation*, which can be considered for any group G :

Example 1.5. Let $V := \mathbb{C}$ and π be the trivial action of G on $V = \mathbb{C}$. We call (V, π) the *trivial representation* of G .

The trivial representation is actually very important; but, of course, we cannot know anything about G only by looking at its trivial representation. The important idea of representation theory is to understand the given group G through its various representations (“what kinds of shadows G can cast on vector spaces”). Accordingly, the most important problem in representation theory is:

classifying all representations of G .

Example 1.6. Let G be a group and (π, V) be a 1-dimensional representation of G . Then π is identified with a homomorphism $\pi: G \rightarrow \mathbb{C}^\times$ (since $\text{Aut}_{\mathbb{C}}(V) = \mathbb{C}^\times$). A homomorphism $G \rightarrow \mathbb{C}^\times$ is often referred to as a *character* (or *1-dimensional character*) of G .

- Remark 1.7.**
- (1) In this course, we only consider representations of a group realized on $\mathbb{C}$ -vector spaces. However, in general, representation theory heavily depends on the coefficient field of vector spaces. Especially, the whole theory drastically changes when we replace $\mathbb{C}$ with a non-algebraically-closed field or a field of positive characteristic (basically, theory becomes much more difficult).

- (2) It is often the case that the group G of our interest is a topological group (e.g., Lie groups, algebraic groups, profinite groups such as infinite Galois groups, etc...). In this case, it is not so meaningful to consider all representations of G which does not take into account the topology of G . If we really want to understand the topological nature of G , it is reasonable to consider, for example, continuous representations of G (i.e., require that $\pi: G \rightarrow \text{Aut}_{\mathbb{C}}(V)$ or $\pi: G \times V \rightarrow V$ are continuous). But then, we also encounter a question “which kind of topology should we put on V ?” If V is a finite-dimensional $\mathbb{C}$ -vector space, then V has a natural topology. However, it is not the case when V is infinite dimensional or the coefficient field is general. In summary, to understand G well, it is important to think about the “appropriate class” of representations of G .

Example 1.8. Let $G := \mathfrak{S}_n$ be the symmetric group consisting of permutations of n letters. By putting $V := \mathbb{C}^{\oplus n} = \mathbb{C}e_1 \oplus \cdots \oplus \mathbb{C}e_n$, we can let $\mathfrak{S}_n$ naturally act on V by permuting $\{e_1, \dots, e_n\}$. The resulting representation is called the *standard representation* of $\mathfrak{S}_n$ on V .

Example 1.9. Let $G := \text{GL}_n(\mathbb{C})$. Then G naturally acts on $V := \mathbb{C}^{\oplus n}$. We call this representation the *standard representation* of $\text{GL}_n(\mathbb{C})$.

1.2. Irreducible representations. Let us think about “classifying all representations” of a given group G . We start with the following notion:

Definition 1.10. Let (π, V) and (π', V') be representations of G .

- (1) A *homomorphism* (or also called *G -homomorphism*) from (π, V) to (π', V') is a $\mathbb{C}$ -vector space homomorphism $f: V \rightarrow V'$ which preserves G -actions, that is, f satisfies $f(\pi(g)(v)) = \pi'(g)(f(v))$ for any $g \in G$ and $v \in V$.
- (2) An *isomorphism* (or also called *G -isomorphism*) from (π, V) to (π', V') is a homomorphism which is bijective. If there exists an isomorphism between (π, V) and (π', V') , we say that “ (π, V) and (π', V') are isomorphic”.

Looking at this definition, it should be reasonable to think that we may identify representations which are isomorphic to each other. So let us think about classifying all isomorphism classes of representations.

Definition 1.11. Let (π, V) be a representation of a group G .

- (1) If a subspace $W \subset V$ is closed under the G -action on V , i.e., we have $\pi(g)(w) \in W$ for any $g \in G$ and $w \in W$, we say that W is a *subrepresentation* of (π, V) .
- (2) Let $W \subset V$ be a subrepresentation of (π, V) . By letting $g \in G$ act on V/W by $g \cdot (v + W) := g \cdot v + W$ for $v \in V$, we obtain a representation of G on V/W . We call V/W the *quotient representation* of (π, V) by W .

Lemma 1.12. Let $f: (\pi, V) \rightarrow (\pi', V')$ be a homomorphism of representations of G . Then we have the following:

- (1) $\text{Ker}(f) \subset V$ is a subrepresentation.
- (2) $\text{Im}(f) \subset V'$ is a subrepresentation.
- (3) f induces an isomorphism $V/\text{Ker}(f) \cong \text{Im}(f)$ of representations of G .

Proof. We first show (1). Let $v \in \text{Ker}(f)$. Our task is to show that $\pi(g)(v) \in \text{Ker}(f)$ for any $g \in G$. In other words, it suffices to show that $f(\pi(g)(v)) = 0$ for any $g \in G$.

Since f is a homomorphism of representations and $v \in \text{Ker}(f)$, we have

$$f(\pi(g)(v)) = \pi'(g)(f(v)) = \pi'(g)(0) = 0.$$

We next show (2). Let $v' \in \text{Im}(f)$. Our task is to show that $\pi'(g)(v') \in \text{Im}(f)$ for any $g \in G$. Since f is a homomorphism of representations, by writing $v' = f(v)$ with some $v \in V$, we have

$$\pi'(g)(v') = \pi'(g)(f(v)) = f(\pi(g)(v)) \in \text{Im}(f).$$

Finally, by the first isomorphism theorem of linear algebra, we have an isomorphism $V/\text{Ker}(f) \cong \text{Im}(f)$ as $\mathbb{C}$ -vector spaces. It can be easily checked that this isomorphism is also an isomorphism of representations of G . $\square$

As this example shows, we can import various notions and results in linear algebra to representation theory.

Definition 1.13. (1) Let (π, V) and (π', V') be representations of G . We define an representation $\pi \oplus \pi'$ of G on $V \oplus V'$ by

$$(\pi \oplus \pi')(g)(v + v') := \pi(g)(v) + \pi'(g)(v')$$

for any $g \in G$, $v \in V$, $v' \in V'$. We call $(\pi \oplus \pi', V \oplus V')$ the *direct sum* of (π, V) and (π', V') .

(2) Let (π, V) and (π', V') be representations of G . We define an representation $\pi \otimes \pi'$ of G on $V \otimes V'$ by

$$(\pi \otimes \pi')(g)(v \otimes v') := \pi(g)(v) \otimes \pi'(g)(v')$$

for any $g \in G$, $v \in V$, $v' \in V'$. We call $(\pi \otimes \pi', V \otimes V')$ the *tensor product* of (π, V) and (π', V') .

(3) Let (π, V) be a representation of G . We define the dual space $V^\vee := \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ with an action $\pi^\vee$ of G by

$$\langle \pi^\vee(g)(v^\vee), v \rangle := \langle v^\vee, \pi(g^{-1})(v) \rangle$$

for any $g \in G$, $v \in V$, $v^\vee \in V^\vee$, where $\langle -, - \rangle$ denotes the natural pairing

$$V^\vee \times V \rightarrow \mathbb{C}: (v^\vee, v) \mapsto v^\vee(v).$$

We call $(\pi^\vee, V^\vee)$ the *dual representation* (or *contragredient representation*) of (π, V) .

In a more sophisticated language, we can say that the category of representations of a group G has a structure of a “tensor category”. Note that, if G is the trivial group $\{1\}$, then a representation of G is just a $\mathbb{C}$ -vector space. Therefore, we may understand that representation theory is a generalization of linear algebra, or linear algebra is the simplest case of representation theory.

Now recall from linear algebra that an important theorem is that “any $\mathbb{C}$ -vector space has a basis”. In other words, this theorem says that any $\mathbb{C}$ -vector space is isomorphic to a direct sum of copies of $\mathbb{C}$ (the trivial representation of the trivial group!). In particular, the isomorphism classes of $\mathbb{C}$ -vector spaces are classified only by their dimensions.

Keeping this in mind, let us think about the classification of isomorphism classes of representations of a general group G . Firstly, we need to introduce “atoms” in representation theory, which cannot be decomposed any further like “ $\mathbb{C}$ ” in linear algebra.

Definition 1.14. Let (π, V) be a representation of G . If $V \neq \{0\}$ and there does not exist a subrepresentation of V other than $\{0\}$ and V , we say that (π, V) is *irreducible*.

Ideally, we can hope to have the following kind of results for representations of a group G , which naturally generalizes the classification of $\mathbb{C}$ -vector spaces in linear algebra:

- (1) Any representation of G is a direct sum of irreducible representations (“*semisimplicity*”).
- (2) We can give a complete classification of the isomorphism classes of irreducible representations of G (“*classification of irreducible representations*”).

If we can really establish such results, we see that the isomorphism classes of representations of G are exactly given by

$$\bigoplus_{\pi} \pi^{\oplus m_{\pi}},$$

where π runs over the isomorphism classes of irreducible representations of G (which can be further classified by (2)) and $m_{\pi} \in \mathbb{Z}_{\geq 0}$ is the *multiplicity* of π in the direct sum decomposition.

However, the reality is not so sweet; both (1) and (2) depend heavily on the given group G . Especially, (1) is not necessarily true, and (2) is a very difficult problem in general.

Now let us explain the highlight of the first part of this course, which is the representation theory of finite groups. The main point of representation theory of finite groups consists of the following:

- Any finite-dimensional representation of a finite group is semisimple, i.e., decomposes into the direct sum of irreducible representations (thus (1) of the above expectation is true).
- If a representation ρ of G and an irreducible representation π of G are given, we can establish a uniform manner of determining the multiplicity of π in the irreducible decomposition of ρ .

Here, we remark that the problem of classifying irreducible representations is still a difficult problem in general, even in the case of finite groups. For example, it is known that irreducible representations of symmetric groups can be classified by Young diagrams, which are very explicit combinatorial objects. However, such a nice classification exists only for very few groups (probably), and basically depends on each individual group. In other words, we can interpret that the “general theory of representation theory of finite groups” is a claim that “classification of all representations can be completely reduced to classification of irreducible representations”.

1.3. Semisimplicity.

Definition 1.15. Let (π, V) be a finite-dimensional representation of G . If the $\mathbb{C}$ -vector space V is equipped with a Hermitian inner product $(-, -): V \times V \rightarrow \mathbb{C}$ which is G -invariant, i.e., the equality

$$(\pi(g)(v), \pi(g)(v')) = (v, v')$$

is satisfied by any $g \in G$ and $v, v' \in V$, we say that (π, V) is a *unitary representation* of G .

Lemma 1.16. *Let G be a group and (π, V) be a unitary finite-dimensional representation of G . Then, for any subrepresentation $W \subset V$, there exists a subrepresentation $W' \subset V$ such that $V = W \oplus W'$.*

Proof. We let $W^\perp$ be the orthogonal complement of W in V with respect to the inner product $(-, -)$, i.e.,

$$W^\perp := \{v \in V \mid (v, w) = 0 \text{ for any } w \in W\}.$$

Then, since $(-, -)$ is an inner product, we have $V = W \oplus W^\perp$. Thus it is enough to check that $W^\perp$ is a subrepresentation. For any $v \in W^\perp$ and $g \in G$, the G -invariance of $(-, -)$ implies that

$$(\pi(g)(v), w) = (v, \pi(g^{-1})w) = 0$$

for any $w \in W$ (note that $\pi(g^{-1})w \in W$ since W is closed under G -action). In other words, $\pi(g)v$ again belongs to $W^\perp$. $\square$

Lemma 1.17. *Any unitary finite-dimensional representation of a group G is semisimple.*

Proof. Let (π, V) be a unitary finite-dimensional representation of G . If V is already irreducible, then π itself gives an irreducible decomposition of π . Thus let us suppose that V is not irreducible in the following. Then, by definition, there exists a non-zero proper subrepresentation $\{0\} \subsetneq W \subsetneq V$. By applying Lemma 1.16 to W , we can choose a subrepresentation $W' \subset V$ such that $V = W \oplus W'$. Since the restriction of the inner product of V to W' gives an inner product of W' , we see that W' is again a unitary finite-dimensional representation of G . By repeating this procedure, we eventually arrive at an irreducible decomposition of V since V is finite-dimensional. $\square$

Proposition 1.18. *Any finite-dimensional representation of a finite group G is semisimple.*

Proof. By Lemma 1.17, it is enough to prove that any finite-dimensional representation of a finite group G is automatically unitary. Let $(-, -): V \times V \rightarrow \mathbb{C}$ be any inner product (which may not necessarily be G -invariant). We modify $(-, -)$ by “averaging” it with respect to the G -action:

$$(-, -)': V \times V \rightarrow \mathbb{C}; \quad (v, v')' := \frac{1}{|G|} \sum_{g \in G} (\pi(g)(v), \pi(g)(v')).$$

This is obviously again an inner product (positive-definite Hermitian inner product) of V . Moreover, it is G -invariant; indeed, for any $h \in G$ and $v, v' \in V$, we have

$$\begin{aligned} (\pi(h)(v), \pi(h)(v'))' &= \frac{1}{|G|} \sum_{g \in G} (\pi(gh)(v), \pi(gh)(v')) \\ &= \frac{1}{|G|} \sum_{g \in G} (\pi(g)(v), \pi(g)(v')) = (v, v')'. \end{aligned}$$

$\square$

1.4. Exercises.

Exercise 1.1. Let G be a finite group. Prove that any irreducible representation of G is necessarily finite-dimensional.

Exercise 1.2. Let $G := \mathbb{Z}$. We consider the 2-dimensional representation (π, V) of G defined by $V = \mathbb{C}^{\oplus 2}$ and

$$\pi: \mathbb{Z} \rightarrow \mathrm{GL}_2(\mathbb{C}): n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

Prove that π is not semisimple.

2. LECTURE 2: CHARACTERS OF FINITE-DIMENSIONAL REPRESENTATIONS

2.1. Characters of representations.

Definition 2.1. Let (π, V) be a finite-dimensional representation of a group G . We define a function $\Theta_\pi: G \rightarrow \mathbb{C}$ by

$$\Theta_\pi: G \rightarrow \mathbb{C}: g \mapsto \text{Tr}(\pi(g) | V).$$

We call Θ_π the *character* (or *trace character*) of (π, V) .

Practically, the trace character can be computed by expressing $\pi(g)$ as a square matrix with respect to any $\mathbb{C}$ -basis of V (recall that the trace of a $\mathbb{C}$ -linear endomorphism is independent of the choice of a $\mathbb{C}$ -basis).

Remark 2.2. Recall that we also call a 1-dimensional representation of G a “character” of G . So these terminologies are confusing, but I believe there will not be a serious problem. (If (π, V) is a 1-dimensional character, then its trace character $\Theta_\pi: G \rightarrow \mathbb{C}$ is nothing but $\pi: G \rightarrow \mathbb{C}^\times$ itself.)

Let us investigate basic properties of characters of representations.

Lemma 2.3. Let (π, V) be an n -dimensional representation of a finite group G . Then we have the following:

- (1) $\Theta_\pi(1) = n$.
- (2) $\Theta_\pi(g^{-1}) = \overline{\Theta_\pi(g)}$ (the right-hand side denotes the complex conjugate).
- (3) Θ_π is conjugation-invariant, i.e., we have $\Theta_\pi(hgh^{-1}) = \Theta_\pi(g)$ for any $g \in G$ and $h \in G$.

Proof. By choosing a $\mathbb{C}$ -basis of V , let us think of π as a homomorphism $G \rightarrow \text{GL}_n(\mathbb{C})$.

Since $\pi(1)$ is the identity matrix I_n of size n , its trace is given by n . Thus we get (1).

We check (2). Let $\alpha_1, \dots, \alpha_n$ be the set of eigenvalues of $\pi(g)$. Since G is a finite group, we have $g^r = 1$ for some $r \in \mathbb{Z}$. Then we have $\pi(g)^r = \pi(g^r) = I_n$. In particular, this implies that all $\alpha_1, \dots, \alpha_n$ are r -th roots of unity. On the other hand, since $\pi(g^{-1}) = \pi(g)^{-1}$, the eigenvalues of $\pi(g^{-1})$ are given by $\alpha_1^{-1}, \dots, \alpha_n^{-1}$. For any root of unity, its inverse is nothing but its complex conjugate. Hence we get

$$\Theta_\pi(g^{-1}) = \alpha_1^{-1} + \dots + \alpha_n^{-1} = \overline{\alpha_1} + \dots + \overline{\alpha_n} = \overline{\alpha_1 + \dots + \alpha_n} = \overline{\Theta_\pi(g)}.$$

Finally, (3) follows from a general property of the trace of a matrix, that is, the trace is invariant under conjugation:

$$\Theta_\pi(hgh^{-1}) = \text{Tr}(\pi(hgh^{-1})) = \text{Tr}(\pi(h)\pi(g)\pi(h)^{-1}) = \text{Tr}(\pi(g)) = \Theta_\pi(g).$$

□

Definition 2.4. For a group G , we call any function $f: G \rightarrow \mathbb{C}$ which is invariant under G -conjugation a *class function on G* . We write $C(G)$ for the set of class functions on G (note that $C(G)$ naturally has a structure of a $\mathbb{C}$ -vector space).

Note that, with this definition, we may understand that Lemma 2.3 (3) asserts that the character Θ_π of any representation π of G belongs to $C(G)$.

Lemma 2.5. Let (π, V) be a finite-dimensional representations of a group G . Then we have $\Theta_\pi(g) = \Theta_{\pi^\vee}(g^{-1})$ for any $g \in G$.

Proof. Recall that, by definition, the contragredient representation $\pi^\vee$ is realized on the dual space $V^\vee$ by

$$\langle \pi^\vee(g)(v^\vee), v \rangle := \langle v^\vee, \pi(g^{-1})(v) \rangle.$$

The assertion immediately follows from this. $\square$

Lemma 2.6. *Let (π_1, V_1) and (π_2, V_2) be finite-dimensional representations of a group G . Then we have $\Theta_{\pi_1 \oplus \pi_2} = \Theta_{\pi_1} + \Theta_{\pi_2}$.*

Proof. This is almost obvious from the definition; if we fix $\mathbb{C}$ -bases of V_1 and V_2 and express $\pi_1(g)$ and $\pi_2(g)$ as square matrices, then $(\pi_1 \oplus \pi_2)(g)$ is expressed as the block matrix with block-diagonal entries $\pi_1(g)$ and $\pi_2(g)$. $\square$

Example 2.7. Let $G := \mathfrak{S}_3 = \{1, (12), (23), (31), (123), (321)\}$. Let $(\pi, V := \mathbb{C}^{\oplus 3})$ be the standard representation of $\mathfrak{S}_3$. Recall that the conjugacy classes of the symmetric groups are classified by the “cycle types”. In the case of $\mathfrak{S}_3$, we have 3 conjugacy classes:

$$\{1\}, \quad \{(12), (23), (31)\}, \quad \{(123), (321)\}.$$

Let us compute the character Θ_π at each conjugacy class.

- (1) If $g = 1$, then we have $\Theta_\pi(g) = \dim(V) = 3$.
- (2) If $g = (12)$, then the representation matrix of $\pi(g)$ is given by

$$\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}.$$

Hence we get $\Theta_\pi(g) = 1$.

- (3) If $g = (123)$, then the representation matrix of $\pi(g)$ is given by

$$\begin{pmatrix} & & 1 \\ 1 & & \\ & 1 & \end{pmatrix}.$$

Hence we get $\Theta_\pi(g) = 0$.

2.2. Schur’s lemma. Let G be a finite group.

Lemma 2.8. *Let (π_1, V_1) and (π_2, V_2) be irreducible representations of G which are not isomorphic. Then $\text{Hom}_G(V_1, V_2) = \{0\}$. In other words, any homomorphism $f: V_1 \rightarrow V_2$ as a representation of G is necessarily the zero map.*

Proof. Let $f: V_1 \rightarrow V_2$ be a homomorphism as representations of G . Then its kernel $\text{Ker}(f)$ is a subrepresentation of V_1 (Lemma 1.12 (1)). Since V_1 is irreducible, we must have either $\text{Ker}(f) = \{0\}$ or $\text{Ker}(f) = V_1$. If $\text{Ker}(f) = V_1$, then it means that $f = 0$. Thus let us suppose that $\text{Ker}(f) = \{0\}$ for the sake of contradiction. By Lemma 1.12 (3), this implies that $V_1 \cong V_1 / \text{Ker}(f) \cong \text{Im}(f) \subset V_2$. However, as $\text{Im}(f)$ is also a subrepresentation of V_2 (Lemma 1.12 (2)) and nonzero, we necessarily have $\text{Im}(f) = V_2$. Accordingly, we get $V_1 \cong V_2$, hence a contradiction. $\square$

Theorem 2.9 (Schur’s lemma). *Let (π, V) be an irreducible finite-dimensional¹ representation of G . Then $\text{End}_G(V) \cong \mathbb{C}$. In other words, any endomorphism $f: V \rightarrow V$ as a representation of G is necessarily of the form $f = \lambda \cdot \text{id}_V$ with $\lambda \in \mathbb{C}$.*

¹This finite-dimensionality condition is in fact unnecessary; see Exercise of Lecture 1.

Proof. Any $\mathbb{C}$ -linear endomorphism of V of the form $\lambda \cdot \text{id}_V$ is obviously an endomorphism of V as a representation of G . Hence our task is to show its converse.

Let $f: V \rightarrow V$ be a nonzero endomorphism of (π, V) as a representation of G . Let $v \in V \setminus \{0\}$ be an eigenvector with eigenvalue $\lambda \in \mathbb{C}$, i.e., $f(v) = \lambda \cdot v$. Then $f - \lambda \cdot \text{id}_V \in \text{End}_{\mathbb{C}}(V)$ again belongs to $\text{End}_G(V)$. Since the kernel of any homomorphism as G -representations is a subrepresentation (Lemma 1.12 (2)), we see that $\text{Ker}(f - \lambda \cdot \text{id}_V) \subset V$ is a subrepresentation. Since V is irreducible and $\text{Ker}(f - \lambda \cdot \text{id}_V) \neq 0$ (as it at least contains $v \neq 0$), we necessarily have $\text{Ker}(f - \lambda \cdot \text{id}_V) = V$. In other words, $f = \lambda \cdot \text{id}_V$. $\square$

Corollary 2.10. *Let (π_1, V_1) and (π_2, V_2) be irreducible representations of a group G . For any $f \in \text{Hom}_{\mathbb{C}}(V_1, V_2)$, we define $f^0 \in \text{Hom}_{\mathbb{C}}(V_1, V_2)$ by*

$$f^0 := \frac{1}{|G|} \sum_{g \in G} \pi_2(g)^{-1} \circ f \circ \pi_1(g).$$

Then we have

$$f^0 = \begin{cases} 0 & \text{if } \pi_1 \not\cong \pi_2, \\ \frac{\text{Tr}(f)}{\dim(V)} \cdot \text{id}_V & \text{if } \pi_1 \cong \pi_2, \end{cases}$$

where, in the second case, we assume that $\pi_1 = \pi_2$ and $V_1 = V_2 =: V$.

Proof. We note that f^0 is a homomorphism as representations of G . Indeed, for any $h \in G$, we have

$$\begin{aligned} f^0 \circ \pi_1(h) &= \frac{1}{|G|} \sum_{g \in G} \pi_2(g)^{-1} \circ f \circ \pi_1(g) \circ \pi_1(h) \\ &= \frac{1}{|G|} \sum_{g \in G} \pi_2(g)^{-1} \circ f \circ \pi_1(gh) \\ &= \frac{1}{|G|} \sum_{g \in G} \pi_2(gh^{-1})^{-1} \circ f \circ \pi_1(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \pi_2(h) \circ \pi_2(g)^{-1} \circ f \circ \pi_1(g) = \pi_2(h) \circ f^0, \end{aligned}$$

where, in the third equality, we replaced gh with g . Therefore, by Lemma 2.8 and Theorem 2.9, we get

$$f^0 = \begin{cases} 0 & \text{if } \pi_1 \not\cong \pi_2, \\ \lambda \cdot \text{id}_V & \text{if } \pi_1 \cong \pi_2, \end{cases}$$

with some scalar $\lambda \in \mathbb{C}$. Thus our remaining task is to determine the scalar λ when $\pi_1 \cong \pi_2$. We may assume $\pi_1 = \pi_2$ and write π for $\pi_1 = \pi_2$. Then

$$\text{Tr}(f^0) = \frac{1}{|G|} \sum_{g \in G} \text{Tr}(\pi(g)^{-1} \circ f \circ \pi(g)) = \frac{1}{|G|} \sum_{g \in G} \text{Tr}(f) = \text{Tr}(f).$$

As $f^0 = \lambda \cdot \text{id}_V$, we also have $\text{Tr}(f^0) = \lambda \cdot \dim(V)$. Hence we get $\lambda = \text{Tr}(f) / \dim(V)$. $\square$

Let us furthermore rewrite this corollary in terms of matrix representations. By fixing $\mathbb{C}$ -bases of V_1 and V_2 , we write $\pi_2(g^{-1})$ and $\pi_1(g)$ as

$$(\pi_2(g^{-1})_{ij})_{ij} \quad \text{and} \quad (\pi_1(g)_{pq})_{pq},$$

respectively.

Corollary 2.11. (1) If $\pi_1 \not\cong \pi_2$, then, for any i, j, p, q ,

$$\sum_{g \in G} \pi_2(g^{-1})_{ij} \cdot \pi_1(g)_{pq} = 0.$$

(2) If $\pi_1 \cong \pi_2$ (assume $\pi_1 = \pi_2$), then, for any i, j, p, q ,

$$\sum_{g \in G} \pi_2(g^{-1})_{ij} \cdot \pi_1(g)_{pq} = \begin{cases} \frac{|G|}{\dim(V)} & \text{if } i = q \text{ and } j = p, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Let us take any $f \in \text{Hom}_{\mathbb{C}}(V_1, V_2)$ and write it as $f = (f_{kl})_{kl}$ using the fixed $\mathbb{C}$ -bases of V_1 and V_2 . Then each entry of the representation matrix $(f_{iq}^0)_{iq}$ of f^0 is given by:

$$f_{iq}^0 = \frac{1}{|G|} \sum_{g \in G} \sum_{j,p} \pi_2(g^{-1})_{ij} \cdot f_{jp} \cdot \pi_1(g)_{pq}.$$

We first consider the case where $\pi_1 \not\cong \pi_2$. In this case, by the above corollary, we have $(f_{iq}^0)_{iq} = 0$ for any $f = (f_{kl})_{kl}$. Hence, the above equality implies that

$$\frac{1}{|G|} \sum_{g \in G} \sum_{j,p} \pi_2(g^{-1})_{ij} \cdot f_{jp} \cdot \pi_1(g)_{pq} = 0$$

for each i, q (for any $f = (f_{kl})_{kl}$). Since f can vary arbitrary, by looking at the coefficients of each f_{jp} on the both hand sides of this equality, we get

$$\sum_{g \in G} \pi_2(g^{-1})_{ij} \cdot \pi_1(g)_{pq} = 0$$

for each i, j, p, q .

We next consider the case where $\pi_1 = \pi_2$. In this case, for any $f = (f_{kl})_{kl}$, f^0 is represented by the scalar matrix with diagonal entries $\text{Tr}(f)/\dim(V)$. In other words, we have

$$\frac{1}{|G|} \sum_{g \in G} \sum_{j,p} \pi_2(g^{-1})_{ij} \cdot f_{jp} \cdot \pi_1(g)_{pq} = \begin{cases} \frac{1}{\dim(V)}(f_{11} + \cdots + f_{nn}) & i = q, \\ 0 & i \neq q, \end{cases}$$

where $n = \dim(V)$. Therefore, we get

$$\sum_{g \in G} \pi_2(g^{-1})_{ij} \cdot \pi_1(g)_{pq} = \begin{cases} \frac{|G|}{\dim(V)} & i = q \text{ and } j = p, \\ 0 & \text{otherwise.} \end{cases}$$

□

2.3. Orthogonality relation. In the following, we continue to assume that G is a finite group. Also, by a “representation of G ”, we always mean a finite-dimensional representation. Let $\mathbb{C}[G]$ denote the space of $\mathbb{C}$ -valued functions on G . We define an inner product $(-, -)_G$ on $\mathbb{C}[G]$ by

$$(f_1, f_2)_G := \frac{1}{|G|} \sum_{g \in G} f_1(g) \cdot \overline{f_2(g)}$$

for any $f_1, f_2 \in \mathbb{C}[G]$. (It is obvious that this indeed defines an inner product; $(-, -)_G$ is $\mathbb{C}$ -linear in f_1 , anti- $\mathbb{C}$ -linear in f_2 , and satisfies $(f, f)_G > 0$ for any nonzero f .) As long as G is fixed and the choice of G is clear from the context, we often simply write $(-, -)$ instead of $(-, -)_G$.

Theorem 2.12. *Let (π_1, V_1) and (π_2, V_2) be irreducible representations of G . Then we have*

$$(\Theta_{\pi_1}, \Theta_{\pi_2}) = \begin{cases} 0 & \text{if } \pi_1 \not\cong \pi_2, \\ 1 & \text{if } \pi_1 \cong \pi_2. \end{cases}$$

Proof. By fixing $\mathbb{C}$ -bases of V_1 and V_2 , we write $\pi_2(g^{-1})$ and $\pi_1(g)$ as

$$(\pi_2(g^{-1})_{ij})_{ij} \quad \text{and} \quad (\pi_1(g)_{pq})_{pq},$$

respectively. Then we have

$$\begin{aligned} (\Theta_{\pi_1}, \Theta_{\pi_2}) &= \frac{1}{|G|} \sum_{g \in G} \Theta_{\pi_1}(g) \cdot \overline{\Theta_{\pi_2}(g)} = \frac{1}{|G|} \sum_{g \in G} \Theta_{\pi_1}(g) \cdot \Theta_{\pi_2}(g^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} \left(\sum_i \pi_1(g)_{ii} \right) \cdot \left(\sum_p \pi_2(g^{-1})_{pp} \right) \end{aligned}$$

If $\pi_1 \not\cong \pi_2$, then Corollary 2.11 (1) immediately implies that this is equal to 0. Thus let us suppose that $\pi_1 = \pi_2$. Again by Corollary 2.11 (2), we get

$$\frac{1}{|G|} \sum_{g \in G} \left(\sum_i \pi_1(g)_{ii} \right) \cdot \left(\sum_p \pi_2(g^{-1})_{pp} \right) = \frac{1}{|G|} \sum_i \frac{|G|}{\dim(V)} = 1.$$

□

Corollary 2.13. *Let (π, V) be a representation of G and let $\pi = \pi_1 \oplus \cdots \oplus \pi_r$ be a decomposition into irreducible representations (recall that there always exists such a decomposition by Proposition 1.18). Then, for any irreducible representation π_0 of G , the number of π_i 's which are isomorphic to π_0 is given by $(\Theta_\pi, \Theta_{\pi_0})$. In particular, a decomposition of any representation G into irreducible representations is always unique up to permutation and isomorphism of irreducible factors.*

Proof. Since $\Theta_\pi = \Theta_{\pi_1} + \cdots + \Theta_{\pi_r}$, we have

$$(\Theta_\pi, \Theta_{\pi_0}) = (\Theta_{\pi_1} + \cdots + \Theta_{\pi_r}, \Theta_{\pi_0}) = \sum_{i=1}^r (\Theta_{\pi_i}, \Theta_{\pi_0}) = |\{1 \leq i \leq r \mid \pi_i \cong \pi_0\}|$$

by Theorem 2.12. □

Corollary 2.14. *Let (π, V) and (π', V') be representations of G . Then $\pi \cong \pi'$ if and only if $\Theta_\pi = \Theta_{\pi'}$.*

Proof. The only if direction is trivial, so let us prove the if direction. Let us take irreducible decompositions of π and π' :

$$\pi \cong \pi_1 \oplus \cdots \oplus \pi_r \quad \text{and} \quad \pi' \cong \pi'_1 \oplus \cdots \oplus \pi'_s.$$

Then, for each irreducible representation π_0 of G , Corollary 2.13 implies that

$$|\{1 \leq i \leq r \mid \pi_i \cong \pi_0\}| = (\Theta_\pi, \Theta_{\pi_0}) = (\Theta_{\pi'}, \Theta_{\pi_0}) = |\{1 \leq j \leq s \mid \pi'_j \cong \pi_0\}|.$$

This shows that the above decompositions of π and π' coincide up to permutations of indices and up to isomorphism. □

At this point, we can clearly understand why the notion of the character of a representation is important in representation theory of finite groups. Any (finite-dimensional) representation of a finite group is uniquely determined by its character, so *characters tell all*.

Another important point is that the character of a representation is a complex-valued function on the group, which is a much more concrete object to work with. Representations have the freedom to “move”: two isomorphic representations can be realized on completely different vector spaces. This freedom makes representation theory so flexible and useful, but, at the same time, also makes things more complicated. For example, when we want to compare two different representations of a group, we first must think about how to relate those two totally different (seemingly) spaces. On the other hand, by passing to their characters, the problem of comparing representations is reduced to the much more concrete problem of comparing complex-valued functions.

Corollary 2.15. *Let π be a representation of G . Then π is irreducible if and only if $(\Theta_\pi, \Theta_\pi) = 1$.*

Proof. We consider the irreducible decomposition

$$\pi \cong \bigoplus_{\pi_0} \pi_0^{\oplus m_{\pi_0}},$$

where the sum runs over isomorphism classes of irreducible representations of G . Then, by Corollary 2.13, we have

$$(\Theta_\pi, \Theta_\pi) = \sum_{\pi_0} (m_{\pi_0} \cdot \Theta_{\pi_0}, m_{\pi_0} \cdot \Theta_{\pi_0}) = \sum_{\pi_0} m_{\pi_0}^2.$$

Hence $(\Theta_\pi, \Theta_\pi) = 1$ if and only if $\sum_{\pi_0} m_{\pi_0}^2 = 1$, which is equivalent to that only one π_0 appears in the irreducible decomposition of π , whose multiplicity $m_{\pi_0} = 1$. $\square$

Let us also present a practical consequence of our discussions so far:

Corollary 2.16. *For any two finite-dimensional representations (π_1, V_1) and (π_2, V_2) of G , we have*

$$\dim \operatorname{Hom}_G(\pi_1, \pi_2) = (\Theta_{\pi_1}, \Theta_{\pi_2}).$$

Proof. By taking irreducible decompositions of π_1 and π_2 , the identity follows from Theorems 2.9 and 2.12. $\square$

Example 2.17. Let us consider the standard representation (π, V) of $G := \mathfrak{S}_3$ again (see Example 2.7). Recall that we have three conjugacy classes of G :

$$\{1\}, \quad \{(12), (23), (31)\}, \quad \{(123), (321)\}.$$

The characters $\Theta_\pi(g)$ on these conjugacy classes are given by 3, 1, 0, respectively. Therefore, we get

$$(\Theta_\pi, \Theta_\pi) = \frac{1}{|\mathfrak{S}_3|} \sum_{g \in \mathfrak{S}_3} \Theta_\pi(g) \cdot \overline{\Theta_\pi(g)} = \frac{1}{6} (1 \cdot 3^2 + 3 \cdot 1^2 + 2 \cdot 0^2) = 2.$$

Hence, at least Corollary 2.15 implies that π is not irreducible. More precisely, if we let $\pi \cong \bigoplus_{\pi_0} \pi_0^{\oplus m_{\pi_0}}$ be the irreducible decomposition, then we have

$$\sum_{\pi_0} m_{\pi_0}^2 = 2.$$

Thus the only possibility is that we have exactly two irreducible representations appearing in the decomposition of π and that the multiplicities are both 1.

Furthermore, if we write $V = \mathbb{C}e_1 \oplus \mathbb{C}e_2 \oplus \mathbb{C}e_3$, then the subspace $\mathbb{C}(e_1 + e_2 + e_3)$ is a subrepresentation of V , on which G acts trivially. Therefore, we see

that π decomposes into the sum of the trivial representation and a 2-dimensional irreducible representation, both of which have multiplicity 1. The 2-dimensional irreducible subrepresentation can be concretely given by $\{x_1e_1 + x_2e_2 + x_3e_3 \in V \mid x_1 + x_2 + x_3 = 0\}$.

2.4. The case of abelian groups.

Proposition 2.18. *Let G be a finite abelian group. Then any irreducible representation of G is 1-dimensional.*

Proof. Let (π, V) be an irreducible representation of G . If we fix $g \in G$, then $\pi(g) \in \text{End}_{\mathbb{C}}(V)$ is in fact an endomorphism as G -representations. Indeed, for any $h \in G$, we have $\pi(g) \circ \pi(h) = \pi(gh) = \pi(hg) = \pi(h) \circ \pi(g)$. By Schur's lemma (Theorem 2.9), this implies that $\pi(g) = z_g \cdot \text{id}_V$ for some $z_g \in \mathbb{C}^\times$. Namely, any $g \in G$ acts on V via a scalar multiplication. Therefore, so that V is irreducible, we necessarily have that V is 1-dimensional. $\square$

2.5. Exercises. In the following exercises, “to describe the irreducible decomposition” means to write a given representation π as

$$\pi \cong \bigoplus_{\pi_0} m_{\pi_0} \pi_0,$$

where the sum is over isomorphism classes of irreducible representations, with explicitly determining each m_{π_0} . (You do not have to find an explicit description of each subspace π_0 in π .)

Exercise 2.1. (1) Let (π_1, V_1) and (π_2, V_2) be finite-dimensional representations of a group G . Prove that $\Theta_{\pi_1 \otimes \pi_2} = \Theta_{\pi_1} \cdot \Theta_{\pi_2}$.

(2) Let (π, V) be the 2-dimensional irreducible representation of $\mathfrak{S}_3$ considered in Example 2.17. Describe the irreducible decomposition of $\pi^{\otimes 2} = \pi \otimes \pi$.

Exercise 2.2. Let X be the set of subsets of $\{1, 2, 3, 4\}$ consisting of two elements:

$$X = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}.$$

Then the permutation group $\mathfrak{S}_4$ naturally acts on X . Accordingly, if we let $V := \bigoplus_{x \in X} \mathbb{C}e_x$ (the $\mathbb{C}$ -vector space with basis labeled by elements of X), we get a representation (π, V) of $\mathfrak{S}_4$. Describe the irreducible decomposition of π .

Exercise 2.3. Consider the dihedral group of order 10:

$$D_{10} := \langle r, s \mid r^5 = s^2 = 1, srs = r^{-1} \rangle.$$

The group D_{10} naturally acts on the set of five vertices of the pentagon. By identifying these vertices with the five standard basis vectors of $\mathbb{C}^5$, we obtain a representation $(\pi, \mathbb{C}^5)$ of D_{10} . Describe the irreducible decomposition of π .

3. LECTURE 3: BASIC RESULTS ON IRREDUCIBLE REPRESENTATIONS

3.1. Square-sum formula for dimensions. Let G be a finite group. Recall that $\mathbb{C}[G]$ denotes the $\mathbb{C}$ -vector space of $\mathbb{C}$ -valued functions on G . We let G act on $\mathbb{C}[G]$ by, for $f \in \mathbb{C}[G]$ and $g \in G$,

$$(R(g) \cdot f)(x) := f(xg).$$

It is not difficult to see that this is indeed a well-defined group action, hence defines a representation $(R, \mathbb{C}[G])$ of G .

Definition 3.1. We call $(R, \mathbb{C}[G])$ the *regular representation* (or *right regular representation*) of G .

The irreducible decomposition of $(R, \mathbb{C}[G])$ has a particularly interesting feature as we will examine from now on.

First, note that $\mathbb{C}[G]$ has the following $\mathbb{C}$ -basis:

$$\{\mathbb{1}_g \mid g \in G\},$$

where $\mathbb{1}_g$ denotes the characteristic function of $\{g\} \subset G$, i.e.,

$$\mathbb{1}_g(x) = \begin{cases} 1 & \text{if } x = g, \\ 0 & \text{if } x \neq g. \end{cases}$$

(Hence, in particular, $\dim \mathbb{C}[G] = |G|$.)

Lemma 3.2. *The character Θ_R of the regular representation R of G is given by*

$$\Theta_R(g) = \begin{cases} |G| & \text{if } g = 1, \\ 0 & \text{if } g \neq 1. \end{cases}$$

Proof. Since R is $|G|$ -dimensional, we have $\Theta_R(1) = \text{Tr}(R(1)) = |G|$.

On the other hand, for any $g \in G$ and $x \in G$, we have

$$R(g)(\mathbb{1}_x) = \mathbb{1}_{xg^{-1}}.$$

Hence, whenever $g \neq 1$, the representation matrix of $R(g)$ with respect to the $\mathbb{C}$ -basis $\{\mathbb{1}_g \mid g \in G\}$ is given by a permutation matrix whose diagonal entries are all zero. In particular, we have $\Theta_R(g) = \text{Tr}(R(g)) = 0$. $\square$

Proposition 3.3. *If (π, V) is an irreducible representation of G , then π is contained in the regular representation R with multiplicity $\dim \pi$. Especially, the irreducible decomposition of R is given by*

$$R \cong \bigoplus_{\pi} \pi^{\oplus \dim \pi},$$

where the sum is over all isomorphism classes of irreducible representations of G .

Proof. Recall that (Corollary 2.13) the multiplicity of π in R is given by the inner product of their characters (Θ_{π}, Θ_R) . By Lemma 3.2,

$$(\Theta_{\pi}, \Theta_R) = \frac{1}{|G|} \sum_{g \in G} \Theta_{\pi}(g) \cdot \overline{\Theta_R(g)} = \frac{1}{|G|} \dim \pi \cdot |G| = \dim \pi.$$

$\square$

Corollary 3.4. *There are only finitely many isomorphism classes of irreducible representations of G . More precisely, we have*

$$\sum_{\pi} (\dim \pi)^2 = |G|,$$

where the sum is over the isomorphism classes of irreducible representations of G .

Proof. By Proposition 3.3, we obtain

$$|G| = \dim R = \sum_{\pi} \dim(\pi^{\oplus \dim \pi}) = \sum_{\pi} (\dim \pi)^2.$$

□

3.2. Number of irreducible representations. Recall that we call an element $f \in \mathbb{C}[G]$ which is G -conjugation-invariant a *class function*. We write $C(G)$ for the $\mathbb{C}$ -subspace of $\mathbb{C}[G]$ of class functions.

$$C(G) := \{f \in \mathbb{C}[G] \mid f \text{ is } G\text{-conjugation-invariant}\} \subset \mathbb{C}[G].$$

Also recall that the character of a representation of G is always a class function. Furthermore, the orthogonality relation (Theorem 2.12) shows that the characters of irreducible representations of G are orthonormal to each other with respect to the inner product on $C(G) \subset \mathbb{C}[G]$. Our aim here is to show the following:

Theorem 3.5. *The characters of irreducible representations of G form an orthonormal basis of $C(G)$. More precisely, if we let $\{\pi_1, \dots, \pi_r\}$ be a set of representatives of isomorphism classes of irreducible representations of G , then $\{\Theta_{\pi_1}, \dots, \Theta_{\pi_r}\}$ is a $\mathbb{C}$ -basis of $C(G)$ satisfying*

$$(\Theta_{\pi_i}, \Theta_{\pi_j}) = \delta_{ij},$$

where δ_{ij} is the Kronecker delta.

Before we prove this theorem, we introduce some notation. Suppose that (π, V) is a representation of G . Then, for any class function $f \in C(G)$, we define an endomorphism $\pi(f) \in \text{End}_{\mathbb{C}}(V)$ by

$$\pi(f) := \sum_{g \in G} f(g) \cdot \pi(g)$$

(“averaging π with weight f ”).

Lemma 3.6. *With the above notation, we have $\pi(f) \in \text{End}_G(V)$, i.e., $\pi(f)$ is an endomorphism of V as a representation of G .*

Proof. It suffices to show that, for any $h \in G$, we have $\pi(f) \circ \pi(h) = \pi(h) \circ \pi(f)$, or equivalently, $\pi(f) = \pi(h) \circ \pi(f) \circ \pi(h)^{-1}$. We have

$$\begin{aligned} \pi(h) \circ \pi(f) \circ \pi(h)^{-1} &= \sum_{g \in G} f(g) \cdot \pi(h) \circ \pi(g) \circ \pi(h)^{-1} \\ &= \sum_{g \in G} f(g) \cdot \pi(hgh^{-1}) \\ &= \sum_{g \in G} f(h^{-1}gh) \cdot \pi(g) = \sum_{g \in G} f(g) \cdot \pi(g) = \pi(f), \end{aligned}$$

where we replaced hgh^{-1} with g in the third equality and used the G -conjugation-invariance of f in the fourth equality. □

Proof of Theorem 3.5. We already saw that $\{\Theta_{\pi_1}, \dots, \Theta_{\pi_r}\}$ is an orthonormal subset. Thus our remaining task is to prove that $\{\Theta_{\pi_1}, \dots, \Theta_{\pi_r}\}$ is a $\mathbb{C}$ -basis of $C(G)$. To show this, it is enough to check that any $f \in C(G)$ satisfying $(f, \Theta_{\pi_i}) = 0$ for any $1 \leq i \leq r$ is necessarily zero.

Let $f \in C(G)$ be such a function. For each irreducible representation (π_i, V_i) , let us compute the endomorphism $\pi_i(f) \in \text{End}_G(V_i)$. By Schur's lemma (Theorem 2.9), $\pi_i(f) = c_i \cdot \text{id}_{V_i}$ with some constant $c_i \in \mathbb{C}$. By the definition of the operator $\pi_i(f)$, we have

$$\text{Tr}(\pi_i(f)) = \sum_{g \in G} f(g) \cdot \text{Tr}(\pi_i(g)) = |G| \cdot (f, \Theta_{\pi_i^\vee})$$

(recall that $\Theta_{\pi_i^\vee}(g) = \overline{\Theta_{\pi_i}(g)}$). On the other hand, we also have $\text{Tr}(\pi_i(f)) = c_i \cdot \dim(V_i)$. This implies that

$$c_i = \frac{|G|}{\dim(V_i)} (f, \Theta_{\pi_i^\vee}).$$

Now we use the assumption on f , i.e., $(f, \Theta_{\pi_i}) = 0$ for all i 's. By noting that $\pi_i^\vee$ is irreducible, we see that $c_i = 0$ for all i 's. Hence, $\pi_i(f) = 0$ for all i 's. Since the regular representation $(R, \mathbb{C}[G])$ of G has the decomposition (Proposition 3.3)

$$R \cong \bigoplus_{\pi_i} \pi_i^{\oplus \dim \pi_i},$$

we also know that $R(f) = 0$. However, by the definition of $R(f)$, we have

$$R(f)(\mathbb{1}_1) = \sum_{g \in G} f(g) \cdot R(g)(\mathbb{1}_1) = \sum_{g \in G} f(g) \cdot (\mathbb{1}_{g^{-1}}).$$

Especially, the equality $R(f)(\mathbb{1}_1) = 0$ implies that $f(g) = 0$ at all $g \in G$. $\square$

Corollary 3.7. *The number of isomorphism classes of irreducible representations of G is equal to the number of conjugacy classes of G .*

Proof. By Theorem 3.5, the number of isomorphism classes of irreducible representations of G is given by $\dim C(G)$. If we let $\{C_1, \dots, C_r\}$ be the set of all conjugacy classes of G , then their characteristic functions

$$\{\mathbb{1}_{C_1}, \dots, \mathbb{1}_{C_r}\}$$

form a $\mathbb{C}$ -basis of $C(G)$. In particular, $\dim C(G)$ is equal to the number of conjugacy classes of G . $\square$

Recall that any irreducible representation of a finite abelian group is 1-dimensional (Proposition 2.18). Using the facts we proved so far, we can upgrade Proposition 2.18 as follows:

Proposition 3.8. *Let G be a finite group. Then G is abelian if and only if any irreducible representation of G is 1-dimensional.*

Proof. Let G be a finite group of order n .

The “only if” part was already proved in Proposition 2.18, but here let us also see how it follows from our discussion in this subsection. Suppose that G is abelian. Then, for any $g \in G$, the conjugacy class containing g is the singleton $\{g\}$:

$$\{xgx^{-1} \in G \mid x \in G\} = \{g\}.$$

Hence we have n conjugacy classes of G . If we let $\pi_1, \dots, \pi_n$ be the irreducible representations, then Corollaries 3.4 and 3.7 imply that

$$n = (\dim \pi_1)^2 + \dots + (\dim \pi_n)^2.$$

Hence we must have $\dim \pi_1 = \dots = \dim \pi_n = 1$.

Let us show the “if” part. Let r be the number of conjugacy classes in G . To show that G is abelian, it suffices to check that $r = n$. Let $\pi_1, \dots, \pi_r$ be the set of irreducible representations of G , which must be all 1-dimensional by assumption. Then Corollary 3.4 implies that

$$n = (\dim \pi_1)^2 + \dots + (\dim \pi_r)^2 = r.$$

This completes the proof. $\square$

Example 3.9. Let us classify the irreducible representations of $\mathfrak{S}_3$. Since $\mathfrak{S}_3$ has three conjugacy classes, we must have three irreducible representations up to isomorphism by Corollary 3.7. Let π_1, π_2, π_3 be three (isomorphism classes of) irreducible representations. By Corollary 3.4, we have

$$|\mathfrak{S}_3| = 6 = (\dim \pi_1)^2 + (\dim \pi_2)^2 + (\dim \pi_3)^2.$$

The only possible triple of positive integers whose square sum equals 6 is $(1, 1, 2)$ up to permutation; thus there are two 1-dimensional irreducible representations and one 2-dimensional representation. These can be explicitly realized as follows:

- The trivial representation is obviously a 1-dimensional irreducible representation of $\mathfrak{S}_3$.
- The sign character $\text{sgn}: \mathfrak{S}_3 \rightarrow \{\pm 1\} \subset \mathbb{C}^\times$ is also a 1-dimensional irreducible representation of $\mathfrak{S}_3$, which is obviously not equal to the trivial representation. Here, recall that sgn is given by

$$\text{sgn}(1) = 1,$$

$$\text{sgn}((12)) = \text{sgn}((23)) = \text{sgn}((31)) = -1,$$

$$\text{sgn}((123)) = \text{sgn}((132)) = 1.$$

- Recall from Example 2.17 that the 3-dimensional standard representation (π, V) of $\mathfrak{S}_3$ decomposes into the sum of two irreducible representations, one of which is the trivial representation (write $\mathbb{1}$). Hence, the quotient representation $\pi/\mathbb{1}$ gives a 2-dimensional irreducible representation of $\mathfrak{S}_3$.

3.3. Exercises.

Exercise 3.1. Let G be the alternating group $\mathfrak{A}_5$ on five letters, i.e.,

$$\mathfrak{A}_5 := \text{Ker}(\text{sgn}: \mathfrak{S}_5 \rightarrow \{\pm 1\}).$$

Determine the dimensions of all irreducible representations of G .

Exercise 3.2. Let $G := \mathfrak{S}_4$. We consider the following 3-dimensional subrepresentation of the permutation representation of G :

$$W := \{(x_1, x_2, x_3, x_4) \in \mathbb{C}^{\oplus 4} \mid x_1 + x_2 + x_3 + x_4 = 0\}.$$

Let $\text{sgn}: \mathfrak{S}_4 \rightarrow \{\pm 1\}$ denote the sign character of G .

- (1) Prove that W and $W \otimes \text{sgn}$ are irreducible and are not isomorphic to each other.
- (2) Determine the dimensions of all irreducible representations of G .

Exercise 3.3. Let G be the quaternion group Q_8 , i.e.,

$$G := Q_8 := \{\pm 1, \pm i, \pm j, \pm k\}, \quad i^2 = j^2 = k^2 = ijk = -1.$$

- (1) Find all 1-dimensional representations of G .
- (2) Determine the dimensions of all irreducible representations of G .

4. LECTURE 4: INDUCED REPRESENTATIONS

4.1. Induced representations. Let G be a group and H its subgroup. If a representation (π, V) of G is given, then we may obtain a representation of H by restricting π to H . The resulting representation is called the *restriction* of (π, V) , and denoted by $(\pi|_H, V)$ or $(\text{Res}_H^G \pi, V)$.

What we want to investigate today is the “converse” of this procedure. That is, let us suppose that a representation (ρ, W) of H is given first. Then, is there any natural way to get a representation of G using it? The answer is yes; it is called the *induction* explained below.

We consider the space $\text{Ind}_H^G W$ of W -valued functions satisfying the “left- (H, ρ) -invariance condition” as follows:

$$\text{Ind}_H^G W := \{f : G \rightarrow W \mid f(hx) = \rho(h)(f(x)) \text{ for any } x \in G, h \in H\}.$$

We define an action $\text{Ind}_H^G \rho$ of G on $\text{Ind}_H^G W$ by the right-translation of W -valued functions, i.e., for any $g \in G$ and $f \in \text{Ind}_H^G W$, we define $g \cdot f \in \text{Ind}_H^G W$ by

$$(g \cdot f)(x) := f(xg).$$

(To be more precise, $g \cdot f$ should be written as $(\text{Ind}_H^G \rho)(g)(f)$, but we avoid it since it is too heavy.) Then we obtain a representation $(\text{Ind}_H^G \rho, \text{Ind}_H^G W)$ of G . We call $(\text{Ind}_H^G \rho, \text{Ind}_H^G W)$ the *induction* of (ρ, W) from H to G .

Example 4.1. Let G be a finite group and $H := \{1\}$. We take a representation (ρ, W) of H to be the trivial representation $(\mathbb{1}, \mathbb{C})$. In this case, the space $\text{Ind}_H^G W$ simply equals the $\mathbb{C}$ -vector space of all $\mathbb{C}$ -valued functions on G , i.e., $\mathbb{C}[G]$. Therefore, the induction $(\text{Ind}_{\{1\}}^G \mathbb{1}, \text{Ind}_{\{1\}}^G \mathbb{C})$ is nothing but the right regular representation of G on $\mathbb{C}[G]$.

Lemma 4.2. We have $\dim(\text{Ind}_H^G W) = [G : H] \cdot \dim W$.

Proof. We put $n := \dim W$ and choose a $\mathbb{C}$ -basis $\{v_1, \dots, v_n\}$ of W . We fix a set of representatives $\{g_1, \dots, g_r\}$ of $H \backslash G$. For each g_i and v_i , we define a W -valued function $f_{i,j} : G \rightarrow W$ by

$$f_{i,j}(g) := \begin{cases} \rho(h)v_i & \text{if } g = hg_i \text{ for some } h \in H, \\ 0 & \text{otherwise.} \end{cases}$$

Then it is easy to see that this is an element of $\text{Ind}_H^G W$ and also that $\{f_{i,j}\}_{i,j}$ forms a $\mathbb{C}$ -basis of $\text{Ind}_H^G W$. In particular, $\dim(\text{Ind}_H^G W) = [G : H] \cdot \dim W$. $\square$

4.2. Frobenius character formula.

Theorem 4.3 (Frobenius character formula). *Let G be a finite group and H a subgroup. Let (ρ, W) be a finite-dimensional representation of H . We put $(\pi, V) := (\text{Ind}_H^G \rho, \text{Ind}_H^G W)$. Then, for any $g \in G$,*

$$\Theta_\pi(g) = \sum_{\substack{x \in H \backslash G \\ xgx^{-1} \in H}} \Theta_\rho(xgx^{-1}).$$

Proof. We use the same notation as in the proof of Lemma 4.2. To be more precise, we fix a set of representatives $\{g_1, \dots, g_r\}$ of $H \backslash G$ and a $\mathbb{C}$ -basis $\{v_1, \dots, v_n\}$ of W to define a $\mathbb{C}$ -basis $\{f_{i,j}\}_{i,j}$ of V . For each $i = 1, \dots, r$, we put

$$V_i := \text{Span}_{\mathbb{C}}\{f_{i,j}\}_j$$

so that

$$V = \bigoplus_{i=1}^r V_i.$$

Note that an element $f \in V$ belongs to V_i if and only if $\text{supp}(f) \subset Hg_i$.

Let $g \in G$. Then, for any $f \in V$, we have $\text{supp}(\pi(g)f) = \text{supp}(f)g^{-1}$. Thus, for each i , if we let i' be the unique index such that $Hg_i g^{-1} = Hg_{i'}$, then $\pi(g)$ maps V_i to $V_{i'}$. Hence,

$$\Theta_\pi(g) = \text{Tr}(\pi(g) | V) = \sum_{\substack{i=1, \dots, r \\ Hg_i g^{-1} = Hg_i}} \text{Tr}(\pi(g) | V_i).$$

Let us compute each summand by fixing i such that $Hg_i g^{-1} = Hg_i$, or equivalently, $g \in g_i^{-1} Hg_i$. For each $j = 1, \dots, n$, we write

$$\pi(g)f_{i,j} = \sum_{l=1}^n a_l^{(j)}(g)f_{i,l}$$

with coefficients $a_l^{(j)}(g) \in \mathbb{C}$. With this notation, $\text{Tr}(\pi(g) | V_i)$ is given by

$$\text{Tr}(\pi(g) | V_i) = \sum_{j=1}^n a_j^{(j)}(g).$$

On the other hand, if we write $g = g_i^{-1} h g_i$ with an element $h \in H$, then, by evaluating the identity $\pi(g)f_{i,j} = \sum_{l=1}^n a_l^{(j)}(g)f_{i,l}$ at g_i , we get

$$f_{i,j}(hg_i) = \sum_{l=1}^n a_l^{(j)}(g)f_{i,l}(g_i),$$

which implies

$$\rho(h)v_j = \sum_{l=1}^n a_l^{(j)}(g)v_l.$$

This identity shows that we also have

$$\text{Tr}(\rho(h) | W) = \sum_{j=1}^n a_j^{(j)}(g).$$

Therefore, in summary, we have obtained

$$\begin{aligned} \Theta_\pi(g) &= \sum_{\substack{i=1, \dots, r \\ Hg_i g^{-1} = Hg_i}} \text{Tr}(\pi(g) | V_i) \\ &= \sum_{\substack{x \in H \backslash G \\ xgx^{-1} \in H}} \text{Tr}(\rho(xgx^{-1}) | W) = \sum_{\substack{x \in H \backslash G \\ xgx^{-1} \in H}} \Theta_\rho(xgx^{-1}). \end{aligned}$$

This completes the proof. $\square$

Corollary 4.4. *Let G be a finite group and H a normal subgroup. Let (ρ, W) be a finite-dimensional representation of H . We put $(\pi, V) := (\text{Ind}_H^G \rho, \text{Ind}_H^G W)$. Then, for any $g \in G$,*

$$\Theta_\pi(g) = \sum_{x \in H \backslash G} \Theta_\rho(xgx^{-1}).$$

4.3. Frobenius reciprocity.

Theorem 4.5. *Let G be a finite group and H its subgroup. Let (π, V) be a finite-dimensional representation of G . Let (ρ, W) be a finite-dimensional representation of H . Then we have*

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \rho) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G \pi, \rho).$$

Proof. We construct $\mathbb{C}$ -linear maps between two homomorphism sets which are inverses to each other.

We first construct a $\mathbb{C}$ -linear map

$$F: \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \rho) \rightarrow \mathrm{Hom}_H(\mathrm{Res}_H^G \pi, \rho).$$

For any $\phi \in \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \rho)$, we define a $\mathbb{C}$ -linear homomorphism $F(\phi): V \rightarrow W$ by

$$F(\phi)(v) := \phi(v)(1)$$

for $v \in V$ (note that $\phi(v)$ is an element of $\mathrm{Ind}_H^G W$, hence a W -valued function on G). Then $F(\phi)$ is also an H -homomorphism. Indeed, using that ϕ is a G -homomorphism, we have

$$F(\phi)(\pi(h)v) = \phi(\pi(h)v)(1) = ((\mathrm{Ind}_H^G \rho)(h)(\phi(v)))(1) = \phi(v)(h).$$

Since $\phi(v)$ is an element of $\mathrm{Ind}_H^G W$, we get

$$\phi(v)(h) = \rho(h)\phi(v)(1) = \rho(h)F(\phi)(v).$$

We next construct a $\mathbb{C}$ -linear map

$$F': \mathrm{Hom}_H(\mathrm{Res}_H^G \pi, \rho) \rightarrow \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \rho).$$

For any $\psi \in \mathrm{Hom}_H(\mathrm{Res}_H^G \pi, \rho)$, we define a $\mathbb{C}$ -linear homomorphism $F'(\psi): V \rightarrow \mathrm{Ind}_H^G W$ by

$$F'(\psi)(v)(x) := \psi(\pi(x)v)$$

for $v \in V$. Then $F'(\psi)$ is a G -homomorphism. Indeed, we have

$$F'(\psi)(\pi(g)v)(x) = \psi(\pi(x)\pi(g)v) = \psi(\pi(xg)v) = F'(\psi)(v)(xg) = ((\mathrm{Ind}_H^G \rho)(g)F'(\psi)(v))(x)$$

(recall that the action $\mathrm{Ind}_H^G \rho$ is defined to be the right-translation for W -valued functions).

It is routine to check that these maps are inverses to each other, so let me omit it. $\square$

Remark 4.6. (1) By Corollary 2.16, we may also interpret Theorem 4.5 as the identity

$$(\Theta_\pi, \Theta_{\mathrm{Ind}_H^G \rho})_G = (\Theta_\pi|_H, \Theta_\rho)_H.$$

(2) In the more sophisticated language of category theory, we may understand that Theorem 4.5 says that the functor

$$\mathrm{Res}_H^G: \mathrm{Rep}(G) \rightarrow \mathrm{Rep}(H)$$

between the categories of finite-dimensional representations of G and H is the “left adjoint” to the functor

$$\mathrm{Ind}_H^G: \mathrm{Rep}(H) \rightarrow \mathrm{Rep}(G).$$

4.4. Exercises.

Exercise 4.1. Let G be a finite group and H a normal subgroup of index 2. Let $\chi: H \rightarrow \mathbb{C}^\times$ be a character of H (1-dimensional representation). Determine the irreducible decomposition of $\text{Ind}_H^G \chi$.

Exercise 4.2. Let $n \in \mathbb{Z}_{>1}$. Prove that $\text{Ind}_{\mathfrak{S}_{n-1}}^{\mathfrak{S}_n} \mathbb{1}$ decomposes into the direct sum of the trivial representation and an $(n-1)$ -dimensional irreducible representation.

APPENDIX A. REGULATIONS FOR EXERCISE SESSIONS AND GRADING

This course has Exercise Sessions once every four lectures. Thus it will be like:

2h lecture $\rightarrow$ 2h lecture $\rightarrow$ 2h lecture $\rightarrow$ 2h Exercise Session
 $\rightarrow$ 2h lecture $\rightarrow$ 2h lecture $\rightarrow$ 2h lecture $\rightarrow$ 2h Exercise Session $\rightarrow \dots$

The grade for this course will be determined by

- (1) written reports and
- (2) oral presentations.

Lecture notes will be uploaded for each lecture, and exercise problems will be included in the notes. Solve these problems and submit your solutions. Basically, you can receive up to **4 points** for each problem. Your final score will be the total number of points you earn. Please collect as many points as you want!

However, I also would like to introduce the following special rules:

- You can receive up to **80 points** in total by submitting written reports.
- If you give an oral presentation of your solution to any problem at the blackboard during an Exercise Session, you can receive up to **20 points**.
- To pass the course, you must give an oral presentation **at least once**.

Example A.1. (1) [oral $\emptyset$] + [written 80 pts] $\rightsquigarrow$ [No credit].
(2) [oral 5 pts] + [written 80 pts] $\rightsquigarrow$ [85 pts].
(3) [oral 15 pts] + [oral 20 pts] + [written 70 pts] $\rightsquigarrow$ [100 pts].
(4) [oral 20 pts] $\times$ 5 + [written $\emptyset$] $\rightsquigarrow$ [100 pts].

The detailed regulations are explained below.

A.1. Written reports.

- (1) Please submit your written solution through NTU COOL or by email to me.
- (2) You may submit your reports at any time during the semester.
- (3) After each Exercise Session, the reports submitted up to that point will be returned to you with your score and brief feedback.
- (4) After getting a feedback, if your score is less than 4, you can modify your solution following the feedback and **re-submit up to once**. If your re-submitted solution is improved, you may get more points.

A.2. Oral presentations.

- (1) During an Exercise Session, students will present solutions problems they chose at the blackboard.
- (2) Each presentation must be given **within 15 minutes**, and then a 5 minutes question/comment time will be given. (So, at maximum, 6 students can give oral presentations in one Exercise Session.)
- (3) Before each Exercise Session, students can express which problem they want to present (through the google spread sheet for this course).
 - The list of students and their preferred problems will be visible to everyone.
 - Each student can select more than one problems.
 - If two or more students wish to present the same problem, priority will be given to the student who has given fewer oral presentations. (If they

have given the same number of oral presentations, the presenter will be selected randomly.)

- Within the day before an Exercise Session, the list of presenters will be announced.
 - I would like to do my best so that presentation opportunities will be allocated to every student equally.
- (4) If you are selected as a presenter, please prepare a 15-minute presentation of your solution. You should try to explain your argument so that the other students, TA (Hao-An Wu), and I can follow it. Please regard this as an opportunity to develop your presentation skills.
- (5) In addition to the oral presentation, **the presenter must submit a written solution** to the problem by the Exercise Session.
- (6) You will receive feedback on your written solution and may revise and re-submit it afterward. An improved written solution may result in a higher score as well. However, the quality of your oral presentation has most weights. For example;
- Suppose that you have prepared your presentation seriously, but during the presentation a mistake in your argument is discovered and the proof breaks down. This is not necessarily a serious problem. If you understand the issue through the subsequent discussion and successfully correct it in your revised written solution, you may still receive a high score.
 - On the other hand, suppose that your written solution is perfect, but it is clear that you have not prepared for the presentation. Similarly, if you are unable to explain your own argument or answer basic questions about your solution, this may result in a low score even if the written solution itself is excellent.

A.3. Q & A.

Question 1. Can I give an oral presentation of a problem which were already presented by someone at the blackboard?

Answer 1. No. Each problem can be presented at the brackboard at most once.

Question 2. Can I submit a written solution to a problem which were already presented by someone at the blackboard?

Answer 2. Yes. In this case, probably you already got to know how to solve the problem. However, what is important in writing a report is not only its correctness, but also how it is organized/explained. Please try to write a report in your words with your understanding even if you already have listened to other student's oral presentation on the problem.

Question 3. Can I submit all written solutions together at the end of the course?

Answer 3. Yes, in principle. However, please keep in mind that it will be more difficult for you to know how many points you can get precisely in advance.

Question 4. Is there any other necessary condition to get the course credit (e.g., attendance)?

Answer 4. No, but please do not forget to give an oral presentation at least once.

REFERENCES

- [Ser77] J.-P. Serre, *Linear representations of finite groups*, Springer-Verlag, New York-Heidelberg, 1977, Translated from the second French edition by Leonard L. Scott, Graduate Texts in Mathematics, Vol. 42.

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